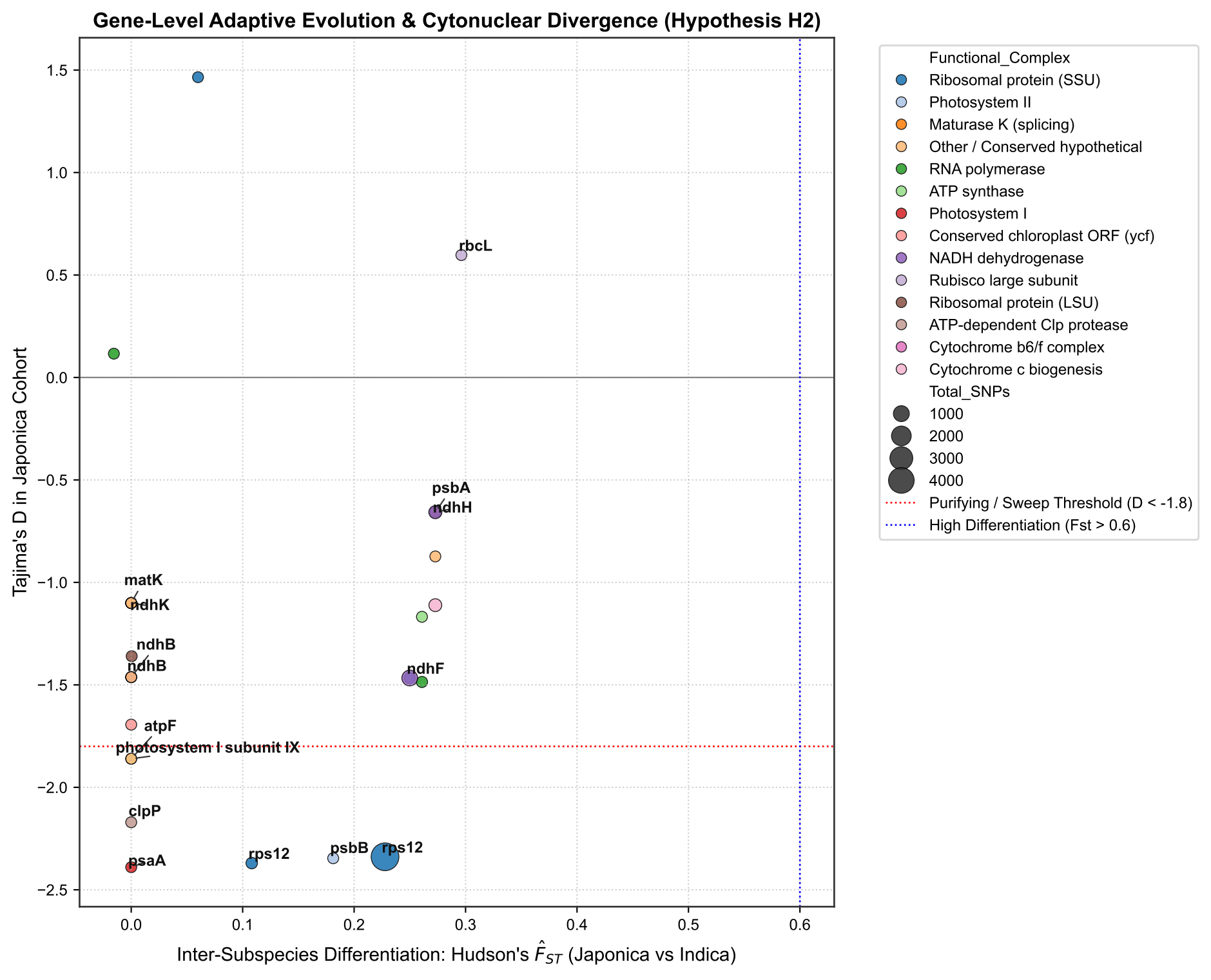
Gene-Level Adaptive Evolution & Cytonuclear Divergence (Hypothesis H2)
Tajima's D in Japonica Cohort
Inter-Subspecies Differentiation: Hudson's $\hat{F}_{ST}$ (Japonica vs Indica)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
0.0
0.1
0.2
0.3
0.4
0.5
0.6
rbcL
psbA
ndhH
matK
ndhK
ndhB
ndhB
ndhF
atpF
photosystem I subunit IX
clpP
psaA
rps12
psbB
rps12
Functional_Complex
Ribosomal protein (SSU)
Photosystem II
Maturase K (splicing)
Other / Conserved hypothetical
RNA polymerase
ATP synthase
Photosystem I
Conserved chloroplast ORF (ycf)
NADH dehydrogenase
Rubisco large subunit
Ribosomal protein (LSU)
ATP-dependent Clp protease
Cytochrome b6/f complex
Cytochrome c biogenesis
Total_SNPs
1000
2000
3000
4000
Purifying / Sweep Threshold (D < -1.8)
High Differentiation (Fst > 0.6)